

Updating the latent posterior $q(x)$ with the drift posterior $q(f)$ fixed

What this note is, in one paragraph

We are fitting a latent stochastic differential equation: a hidden trajectory $x_0, \dots, x_T$ evolves under an unknown drift f , and we see only noisy observations y . Inference is variational EM. We keep a Gaussian approximation $q(x)$ to the hidden path and a Gaussian process posterior $q(f)$ to the drift, and we improve them in alternation. This note covers exactly one of those alternating updates: **holding the drift posterior $q(f)$ fixed, improve the path posterior $q(x)$** . The punchline is that this update is one *exact* rewrite followed by one *single* approximation — and that one approximation, applied once, determines every quantity the update needs.

The whole update in three moves.

M1 (exact). Rewrite the objective so that all of its dependence on the drift f funnels through just *three numbers* per time step — averages of the drift and its slope over $q(x_i)$.

M2 (bookkeeping). We compute those three numbers fast, in a way that hides them from automatic differentiation. So we must tell the optimiser, by hand, how each of them changes with $q(x)$.

M3 (the one approximation). To get those derivatives we make a single assumption — *the drift is a straight line across the width of $q(x_i)$* — and turn the crank. Every derivative the update needs falls out of that one assumption.

1 Setup, and what a $q(x)$ step must compute

The model. Time is discretised (Euler–Maruyama), so between samples the latent state takes a noisy step driven by the drift:

$$x_{i+1} = x_i + \Delta_i f(x_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \Delta_i \Sigma). \quad (1)$$

The variational posterior over the path is a Gaussian Markov chain: each state has a marginal $q(x_i) = \mathcal{N}(m_i, S_i)$, and neighbours share a cross-covariance $S_{i,i+1} = \text{Cov}(x_i, x_{i+1})$. The drift posterior $q(f)$ is already in hand from the other EM update; all we use from it is the posterior-mean drift, written $\bar{f}(x)$. (In the EFGP basis $\bar{f}_r(x) = \phi(x)^* \mu_r$, but nothing below needs that form.)

Why we only ever compute one gradient. We improve $q(x)$ by natural-gradient ascent on the ELBO. For a Gaussian posterior there is a useful shortcut: writing $q(x)$ in its *mean parameters*

$$\mu = (\mathbb{E}[x_i], \mathbb{E}[x_i x_i^\top], \mathbb{E}[x_i x_{i+1}^\top]), \quad (2)$$

the natural gradient is just the *ordinary* gradient in μ ,

$$\tilde{\nabla} \mathcal{L} = \partial \mathcal{L} / \partial \mu. \quad (3)$$

No Fisher matrix, no special calculus: we plainly differentiate the ELBO with respect to μ , damp the result, and step. That machinery — the damping, the conversion back to $(m_i, S_i, S_{i,i+1})$ — is the SING backbone and is not our concern here. **Our only job is to hand the backbone $\partial \mathcal{L} / \partial \mu$.**

Only one piece of the ELBO involves the drift. The ELBO splits into four groups,

$$\mathcal{L} = \underbrace{\mathbb{E}_{q(x)}[\log p(y | x)]}_{\text{observations}} + \sum_i \underbrace{\mathbb{E}_{q(x,f)}[\log p(x_{i+1} | x_i, f)]}_{\text{transitions}} - \underbrace{\text{KL}[q(f) || p(f)]}_{\text{no } q(x)} + \underbrace{H[q(x)] + \mathbb{E}[\log p(x_0)]}_{\text{entropy, initial}}. \quad (4)$$

The observation, entropy, and initial terms are ordinary Gaussian expressions in $q(x)$; the backbone differentiates them directly. The KL does not involve $q(x)$ at all. **Only the transition term couples $q(x)$ to the drift**, so it is the only term we have to work on. Everything from here concerns a single transition i ; call its contribution to the objective trm_i .

2 M1: reduce the transition term to three drift moments (exact)

Start from the log-transition of (1) and take its average under the drift posterior $q(f)$. Two clean things happen. First, the average of f is the posterior mean $\bar{f}$. Second, the average of $f^\top \Sigma^{-1} f$ is $\bar{f}^\top \Sigma^{-1} \bar{f}$ plus a variance term V (the drift's own uncertainty); we carry V along and dispose of it in M3. Writing $\delta_i := x_{i+1} - x_i$,

$$\mathbb{E}_{q(f)}[\log p(x_{i+1} | x_i, f)] = -\frac{\delta_i^\top \Sigma^{-1} \delta_i}{2\Delta_i} + \delta_i^\top \Sigma^{-1} \bar{f}(x_i) - \frac{\Delta_i}{2} \left[\bar{f}(x_i)^\top \Sigma^{-1} \bar{f}(x_i) + V(x_i) \right] + \text{const.} \quad (5)$$

Now average over the path, $\mathbb{E}_{q(x_i, x_{i+1})}[\cdot]$. Two exact simplifications leave the whole thing depending on $q(x)$ through only a handful of quantities.

The first term is already elementary. $\mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \delta_i]$ is a Gaussian average of a quadratic, so it expands into traces of $\mathbb{E}[x_i x_i^\top]$, $\mathbb{E}[x_{i+1} x_{i+1}^\top]$, $\mathbb{E}[x_i x_{i+1}^\top]$ and the means. It is *linear* in μ : no drift, no approximation, and its μ -gradient is immediate.

The middle term is collapsed by Stein's lemma. The cross term $\mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \bar{f}(x_i)]$ couples x_i and x_{i+1} . Stein's lemma trades that coupling for two *marginal* averages against known $q(x)$ statistics:

$$\mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \bar{f}(x_i)] = d_i^\top \Sigma^{-1} \mathbb{E}_{q(x_i)}[\bar{f}] + \text{tr}[\Sigma^{-1} \mathbb{E}_{q(x_i)}[J_{\bar{f}}] C_i^\top], \quad \begin{aligned} d_i &:= m_{i+1} - m_i, \\ C_i &:= S_{i,i+1} - S_i, \end{aligned} \quad (6)$$

where $J_{\bar{f}}$ is the Jacobian (the slope) of the mean drift.

What remains: three drift moments. After (5)–(6), the transition term is a plain algebraic formula in $(m_i, S_i, S_{i,i+1})$ and just three averages of the drift over the marginal $q(x_i)$:

$$\boxed{\begin{aligned} \text{Ef}_i &:= \mathbb{E}_{q(x_i)}[\bar{f}] && (\text{mean drift}), & \text{Edfdx}_i &:= \mathbb{E}_{q(x_i)}[J_{\bar{f}}] && (\text{mean slope}), \\ \text{Eff}_i &:= \mathbb{E}_{q(x_i)}[\bar{f}^\top \Sigma^{-1} \bar{f}] && (\text{mean square}). \end{aligned}} \quad (7)$$

Concretely, dropping the constant and the overall $-1/(2\Delta_i\sigma^2)$ prefactor, one transition contributes

$$\text{trm}_i \propto \mathbb{E}_{q(x_i, x_{i+1})}[\delta_i^\top \Sigma^{-1} \delta_i] + \Delta_i^2 \text{Eff}_i - 2\Delta_i \left(d_i^\top \Sigma^{-1} \text{Ef}_i + \text{tr}[\Sigma^{-1} \text{Edfdx}_i C_i^\top] \right). \quad (8)$$

This is M1, and it is exact. It is also the entire interface between the drift model and the path update: give the backbone the three numbers in (7), and it assembles (8) knowing nothing about where they came from. This mirrors the other EM update, where uncertainty in the path enters the drift fit only by replacing fixed sufficient statistics with their $q(x)$ -averages; here it is the same idea, read the other way.

3 M2: the moments' derivatives must be supplied by hand

We need $\partial \text{trm}_i / \partial \mu$. The explicit algebra of (8) — the δ -quadratic and the d_i, C_i pieces — differentiates by ordinary autodiff, exactly. The difficulty is that the three moments (7) are *themselves* functions of (m_i, S_i) , because they are averages over $q(x_i)$. So the chain rule also demands their derivatives:

$$\frac{\partial \text{Ef}_i}{\partial(m_i, S_i)}, \quad \frac{\partial \text{Edfdx}_i}{\partial(m_i, S_i)}, \quad \frac{\partial \text{Eff}_i}{\partial(m_i, S_i)} \quad (\text{six derivatives}). \quad (9)$$

Why autodiff cannot just do it. Each moment has a closed form (a Gaussian average of a complex exponential), and for speed we evaluate all T of them together in *one* batched transform. But a derivative is needed *per transition*. Asking autodiff to differentiate that batched transform in place would pay one full transform *per transition* instead of one for all of them — orders of magnitude slower. So we compute the moments off the autodiff graph, pass them in as constants, and attach the six derivatives by hand (a `custom_vjp` in JAX). Autodiff then differentiates (8) normally and picks up our injected derivatives through the chain rule.

What the exact derivatives are. For a Gaussian $q(x_i) = \mathcal{N}(m_i, S_i)$, the derivatives of an average are given by Price's theorem (its mean case is Bonnet's lemma):

$$\partial_{m_i} \mathbb{E}_{q(x_i)}[g] = \mathbb{E}_{q(x_i)}[\nabla g], \quad \partial_{S_i} \mathbb{E}_{q(x_i)}[g] = \frac{1}{2} \mathbb{E}_{q(x_i)}[\nabla^2 g]. \quad (10)$$

That is: differentiating in the mean pulls down a *gradient* of the integrand; differentiating in the covariance pulls down a *Hessian*. Applied to the three moments, the six exact answers are

moment	exact $\partial/\partial m_i$	exact $\partial/\partial S_i$
$\mathbb{E}_{q(x_i)}[\bar{f}]$	$\mathbb{E}_{q(x_i)}[J_{\bar{f}}]$	$\frac{1}{2} \mathbb{E}_{q(x_i)}[H_{\bar{f}}]$
$\mathbb{E}_{q(x_i)}[J_{\bar{f}}]$	$\mathbb{E}_{q(x_i)}[H_{\bar{f}}]$	$\frac{1}{2} \mathbb{E}_{q(x_i)}[\nabla^2 J_{\bar{f}}]$
$\mathbb{E}_{q(x_i)}[\bar{f}^\top \Sigma^{-1} \bar{f}]$	$2 \mathbb{E}_{q(x_i)}[J_{\bar{f}}^\top \Sigma^{-1} \bar{f}]$	$\mathbb{E}_{q(x_i)}[J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}}] + \sum_r \sigma_r^{-2} \mathbb{E}_{q(x_i)}[\bar{f}_r H_{\bar{f}_r}]$

where $H_{\bar{f}}$ is the Hessian (the curvature) of the mean drift. Five of the six answers need this curvature; only the top-left — the slope $\mathbb{E}_{q(x_i)}[J_{\bar{f}}]$, which we already have — does not. Computing the curvature everywhere is the expense we would like to avoid. M3 avoids it.

4 M3: one approximation, and the whole table falls out

Here is the single assumption behind the entire $q(x)$ update:

Across the width of $q(x_i)$, replace the drift by a straight line.

We fix the line by *statistical linearisation*: the affine map that matches $\bar{f}$'s first two moments under $q(x_i)$,

$$\ell(x) = a + B(x - m_i), \quad a := \mathbb{E}_{q(x_i)}[\bar{f}], \quad B := \mathbb{E}_{q(x_i)}[J_{\bar{f}}], \quad (11)$$

its offset the mean drift and its slope the *expected* Jacobian. This is not the Taylor tangent at m_i : by Stein's lemma — the identity already used in (6) — $\mathbb{E}_{q(x_i)}[J_{\bar{f}}]$ equals the least-squares slope $\text{Cov}(\bar{f}, x) S_i^{-1}$, so ℓ is the best affine fit to $\bar{f}$ over the whole blob (statistical linearisation; Särkkä, 2013; Oppé & Archambeau, 2009). Both coefficients a, B are the exact gmix moments of M1; the slope is $\mathbb{E}_{q(x_i)}[J_{\bar{f}}]$ throughout, *never* a point value $J_{\bar{f}}(m_i)$. We use ℓ in place of $\bar{f}$ inside every $q(x_i)$ -average; the six derivatives then come out with no further choices.

How to picture it. The assumption is not that the drift is simple — we linearise afresh at *every* m_i , so the tangents track the true curved field along the path. It is a statement about $q(x)$: each marginal $q(x_i)$ is narrow enough that the drift is effectively straight across that one blob. Take a one-dimensional bistable drift, $f(x) = x - x^3$. Over a Gaussian $q(x_i) = \mathcal{N}(m_i, S_i)$ of width $\sqrt{S_i}$ we pretend f is a straight line (of slope $\mathbb{E}_{q(x_i)}[f']$; the point tangent $f'(m_i) = 1 - 3m_i^2$ differs from it only at the order below). The mismatch is set by the curvature, $f'' = -6x$: the error in any averaged drift quantity is of order $\frac{1}{2}f'' S_i$, i.e. (curvature) $\times$ (width). It is zero if f is genuinely affine (a linear SDE, a line attractor), and small whenever $q(x_i)$ is narrow *relative to the scale on which the drift bends* — the length-scale ℓ — so the honest condition is $\|S_i\| \ll \ell^2$, not $\|S_i\|$ small in absolute terms. We expand at the mean precisely because there the first-order error cancels, leaving only this second-order term.

When it breaks. The error is largest where a *wide* blob sits on a *sharply bending* part of the field — both factors at once. In practice that means: the cold-start iterations, before the smoother has localised $q(x)$; stretches of the trajectory with few or noisy observations, where $q(x_i)$ stays broad; and sharp turns of the flow (a fast reversal, a separatrix) where $\|H_{\bar{f}}\|$ is large. A tight blob on a gentle part of the field, or any blob on a near-linear part, is fine.

Why the breaking case does not hurt much. A wide $q(x_i)$ is, by definition, a latent state the data do not pin down. A slightly mis-directed gradient on a weakly-determined coordinate has little leverage on the reconstruction, and SING's ρ -damping attenuates any single overconfident step on top of that — so the approximation is worst exactly where the answer is least sensitive to it. It is also self-correcting: in well-observed regions the update tightens $q(x_i)$, which makes the next linearisation better, so error shrinks over EM iterations rather than accumulating. And the damage is bounded even in the transient: the moment *values* fed to the backbone are the true Gaussian averages of (7) (only their *derivatives* assume a straight line), so the step's base point stays exact and only its *direction* is affected; and each transition's error is local to its own term in the sum, not compounded along the path. (These are the same regimes flagged below for optionally restoring the V term.)

And the worst case is not one we could trust anyway. Sharpen the failure condition: the error is order-one only once the positional standard deviation approaches the length-scale, $\sqrt{\|S_i\|} \gtrsim \ell$. But ℓ is the scale on which the drift varies, and the only information we have about the drift at that scale comes from watching x move between states we have localised to precision $\sqrt{\|S_i\|}$. If $\sqrt{\|S_i\|} \gg \ell$, each look at the drift is a smear wider than the feature it would resolve, so the data cannot have identified structure at scale ℓ in the first place: a confident $\hat{\ell}$ far below the positional spread is simply unsupported. Where the linearisation is worst, the length-scale it conditions on is itself untrustworthy — the two degrade together, and neither is a regime one would report a fit from. Conversely, any well-identified converged fit has $\sqrt{\|S_i\|} \ll \ell$, the path pinned to well within the drift’s variation scale, which is exactly where linearisation is excellent; validity and convergence coincide. (This is the typical, whole-trajectory statement — a few wide- S_i transitions in an otherwise sharp fit are just the weakly-identified states above. And since only $\text{tr}(H_{\bar{f}} S_i)$ enters, uncertainty along directions in which the drift is flat costs nothing, so the true condition is even more forgiving than $\|S_i\| < \ell^2$.)

The six gradients. ℓ is affine, so its moments under $q(x_i) = \mathcal{N}(m, S)$ are exact (the linear part averages out; the Jacobian is the constant B ; the quadratic adds one trace),

$$\mathbb{E}_{q(x_i)}[\ell] = a, \quad \mathbb{E}_{q(x_i)}[J_\ell] = B, \quad \mathbb{E}_{q(x_i)}[\ell^\top \Sigma^{-1} \ell] = a^\top \Sigma^{-1} a + \text{tr}(B^\top \Sigma^{-1} B S),$$

with the m -dependence carried only by $\mathbb{E}_{q(x_i)}[\ell] = a + B(m - m_i)$. Differentiating in (m, S) at $m = m_i$ gives the six injected gradients:

$$\begin{aligned} \partial_m \mathbb{E}_{q(x_i)}[\bar{f}] &= B = \mathbb{E}_{q(x_i)}[J_{\bar{f}}], & \partial_S \mathbb{E}_{q(x_i)}[\bar{f}] &= 0, \\ \partial_m \mathbb{E}_{q(x_i)}[J_{\bar{f}}] &= 0, & \partial_S \mathbb{E}_{q(x_i)}[J_{\bar{f}}] &= 0, \\ \partial_m \mathbb{E}_{q(x_i)}[\bar{f}^\top \Sigma^{-1} \bar{f}] &= 2B^\top \Sigma^{-1} a, & \partial_S \mathbb{E}_{q(x_i)}[\bar{f}^\top \Sigma^{-1} \bar{f}] &= B^\top \Sigma^{-1} B \succeq 0, \end{aligned}$$

$a = \mathbb{E}_{q(x_i)}[\bar{f}]$, $B = \mathbb{E}_{q(x_i)}[J_{\bar{f}}]$ (11), both exact gmix moments. The top-left is the exact Price gradient $\partial_m \mathbb{E}_{q(x_i)}[\bar{f}] = \mathbb{E}_{q(x_i)}[J_{\bar{f}}]$; the bottom-right $B^\top \Sigma^{-1} B \succeq 0$ is the Gauss–Newton curvature of the sum of squares $\bar{f}^\top \Sigma^{-1} \bar{f}$ (Khan & Lin, 2017), keeping the precision update (3) PSD.

The drift variance V : one more term of the same kind, dropped by default. $V_r(x) = \sigma_r^{-2} \phi(x)^* A_r^{-1} \phi(x)$ is a quadratic form in the Fourier features — like the mean-square $\bar{f}_r^2 = \phi^* \mu_r \mu_r^* \phi$ — so it too is a sum of difference-frequency modes $\sum_\delta \omega_r(\delta) e^{2\pi i \delta^\top x}$, and its $q(x_i)$ -average is the *same* characteristic-function envelope-NUFFT as every other term (value and m -gradient exact, S -slot dropped). The dense A_r^{-1} only sets the coefficients $\omega_r(\delta)$: (i) compute the drift posterior variance by any method — Hutchinson on A_r^{-1} , or the pathwise/Matheron estimator we use for UQ — then (ii) apply the same gmix envelope. Production omits V altogether: $\nabla V \approx 0$ wherever the data surround x evenly at scale ℓ ($\|\nabla \mathbb{E}_{q(x_i)}[V]\| \approx 5 \times 10^{-5} \|\partial_m \text{Eff}\|$ on our benchmark, recovery unchanged to 3–4 digits). Restore it (opt-in) when that flatness fails — clustered data, sharp boundary turns, or cold start.

5 One inner iteration, end to end

1. From the current $q(x)$, read off $m_i, S_i, S_{i,i+1}$ and form $d_i = m_{i+1} - m_i$, $C_i = S_{i,i+1} - S_i$.
2. Update the drift posterior $q(f)$ (the other EM step), giving the mean drift $\bar{f}$.

3. **[M1]** Evaluate the three moments (7) at all transitions in one batched transform.
4. **[M2, M3]** Hand them to the backbone as constants, with the linearised derivative table of §4 attached.
5. Autodiff the assembled scalar (8) in μ ; by (3) this is the natural gradient. Damp and step.
6. Repeat for n_{estep} inner iterations, then take the kernel (hyperparameter) M-step.

Crosswalk to the long note

here	<code>efgp_estep.tex</code>
§1 setup, what the step computes	§2 (interface), §6 setup
§2 M1, three drift moments (exact)	§6 “the transition term”, items (i)–(ii)
§3 M2, hand-supplied derivatives	§6 “why we can’t just autodiff”, App. C
§4 M3, the one approximation	§6 Gauss–Newton framework, items (iii)–(iv)
§4 the V drop	§6 item (iv); App. B (restoration)
how the moments are computed	§4 (three backends), <code>efgp_estep_charfun.tex</code>